Generalized Linear Models

Poisson Regression

Linear Regression - Continuous Target

$Y|\vec{x}$ follows a distribution with mean

$$\mu = \beta_0 + \beta_1 x_1 + \cdots + \beta_n x_n$$

Linear Regression - Continuous Target

$Y|\vec{x}$ follows a **normal** distribution with mean

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Logistic Regression - Binary Target

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$$\mu = \qquad (\beta_0 + \beta_1 x_1 + \cdots + \beta_n x_n)$$

Logistic Regression - Binary Target

$Y|\vec{x}$ follows a **Bernoulli** distribution with mean

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Logistic Regression - Binary Target

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$$\mu = \text{logistic}(\beta_0 + \beta_1 x_1 + \cdots + \beta_n x_n)$$

$$= \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \cdots + \beta_n x_n)}}$$

What if our target variable is a count?

Example: Here, our target is the number of times an individual visited the doctor in the last two weeks.

	visits	gender	age	income	illness	rec
1	0	male	22	0.15	1	
2	0	female	52	1.10	1	
3	0	male	72	0.35	0	
4	0	female	19	0.65	2	
5	0	female	47	0.55	5	
6	1	male	19	0.15	1	
7	0	male	47	0.45	0	
8	0	male	22	0.15	1	
9	0	male	22	0.55	1	
10	1	female	52	0.75	1	

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Since this is count data, we can try using a **Poisson distribution** to model the target variable.

We just need to estimate the mean of this distribution.

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We'll estimate the number of visits based on the *age* variable.

Example: Estimate using the *age* variable.

```
pr_hosp <- glm('visits ~ age',
               data = doctor_visits,
               family = "poisson")
```

Example: Estimate using the *age* variable.

```
summary(pr_hosp)
```

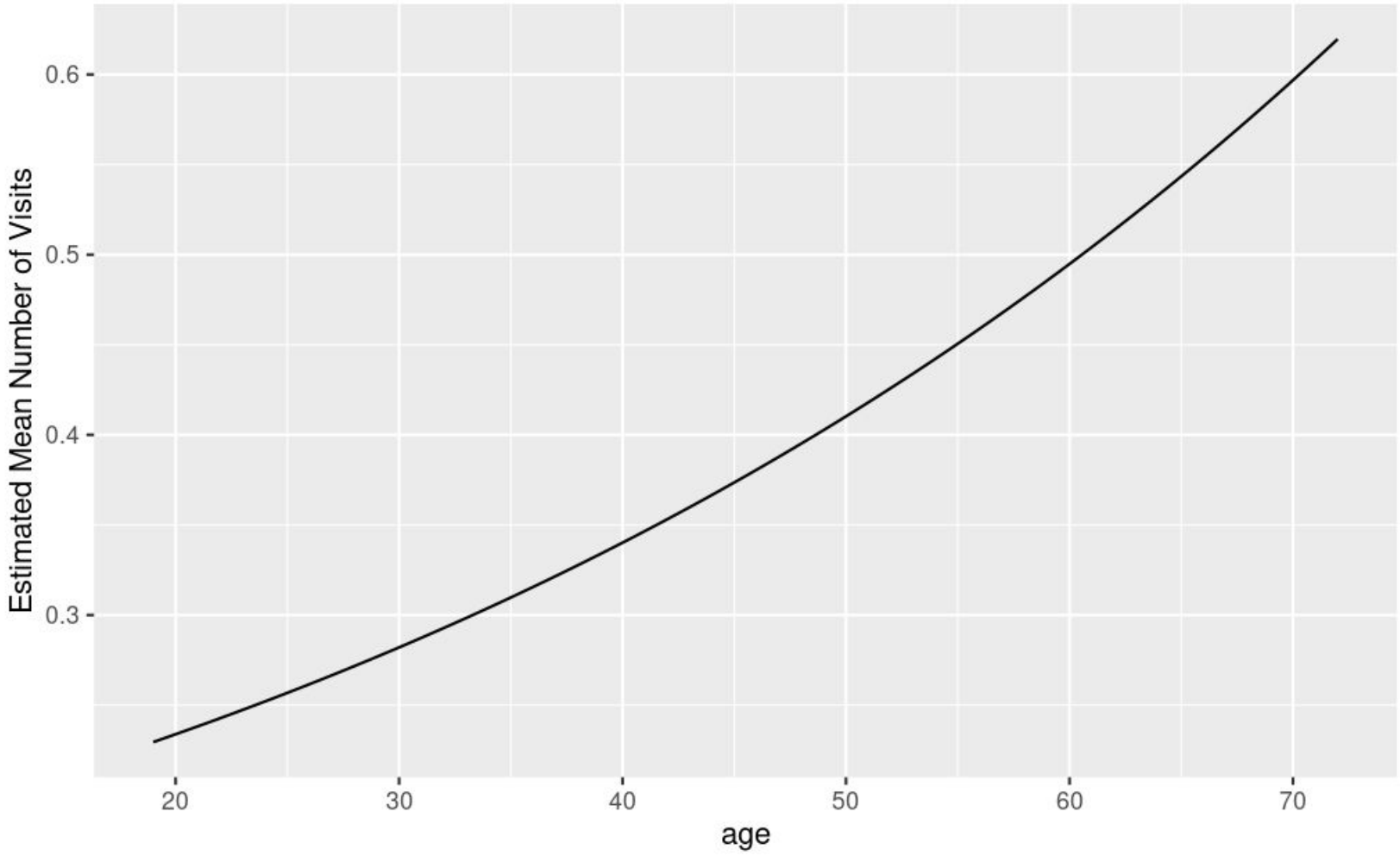
```
Call:
glm(formula = "visits ~ age", family = "poisson", data = doctor_visits)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -1.82797    0.44123  -4.143 3.43e-05 ***
age          0.01874    0.00782   2.396   0.0166 *
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

For a person whose age is *t*, the estimated value of the mean is

$$\exp(-1.8280 + 0.01874t)$$

$$= e^{(-1.8280 + 0.01874t)}$$


Estimated Mean Number of Visits
0.6
0.5
0.4
0.3
20
30
40
50
60
70
age

Summary - Linear, Logistic, and Poisson Regression

Linear Regression

$Y|\vec{x}$ follows a **normal** distribution with mean

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$$= \frac{1}{1 + e^{-(\beta_0 + \beta_1 x_1 + \cdots + \beta_n x_n)}}$$

Poisson Regression

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$$= e^{\beta_0 + \beta_1 x_1 + \cdots + \beta_n x_n}$$

Example: MLB Data

	runs	games	OBP
1	582	139	0.310
2	671	137	0.320
3	566	137	0.304
4	716	141	0.328
5	602	140	0.306
6	692	138	0.334
7	692	140	0.330
8	606	136	0.306
9	634	139	0.315
10	608	140	0.307

Goal: Estimate the number of runs scored (*runs*) based on the on-base percentage (*OBP*).

Problem: Teams have played a different numbers of games, so the comparison is not really fair.

Potential Fix: Estimate the number of runs scored *per game*.

$$\frac{runs}{games} = e^{\beta_0 + \beta_1 (OBP)}$$

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$$\Longrightarrow \quad \log(runs) = \beta_0 + \beta_1(OBP) + \boxed{\log(games)}$$

This is called an
offset term.

```
pr_mlb <- glm("runs ~ OBP",
              data = mlb,
              family = "poisson",
              offset = log(games) )
```

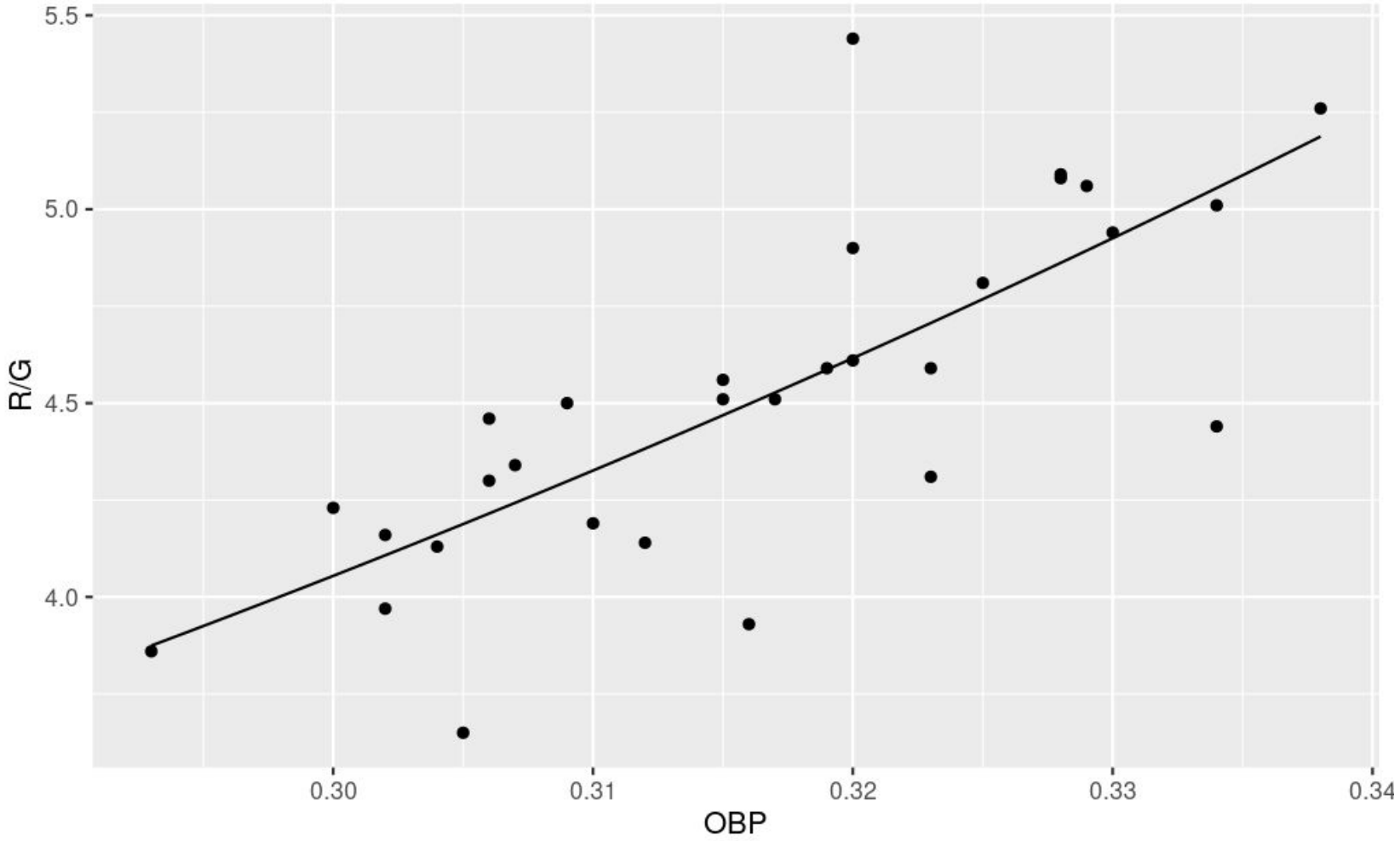
Specify the offset column.

```
summary(pr_mlb)
```

```
Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -0.5456     0.2049  -2.663  0.00774 **
OBP           6.4849     0.6455  10.046  < 2e-16 ***
```

Given the OBP of a team, the model estimates the mean runs per game as

$$= e^{(-0.5456 + 6.4849(OBP))}$$


5.5
5.0
4.5
4.0
R/G
0.30
0.31
0.32
0.33
0.34
OBP